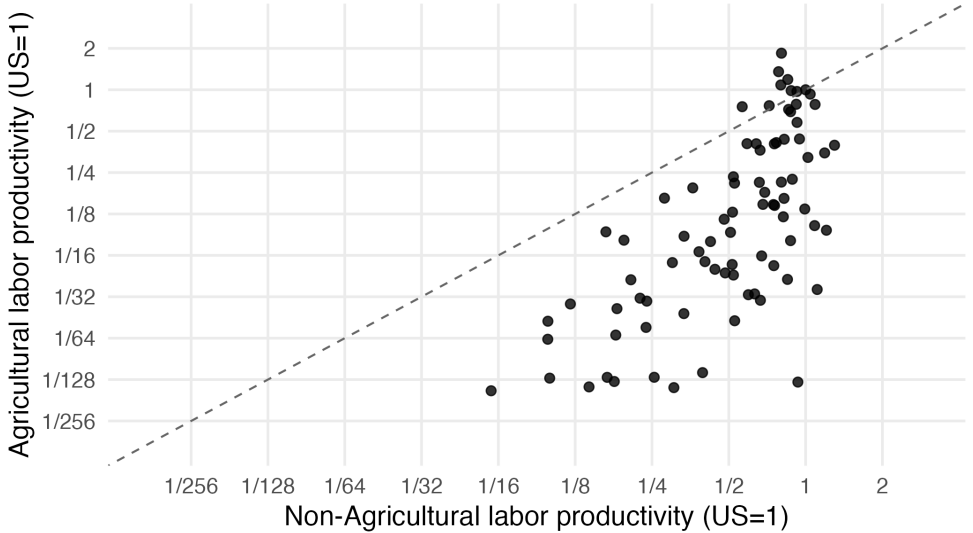
Not Worth Innovating For?
Dynamic Lock-In of Agricultural Productivity Differences

Kazunari Tanabe

LSE

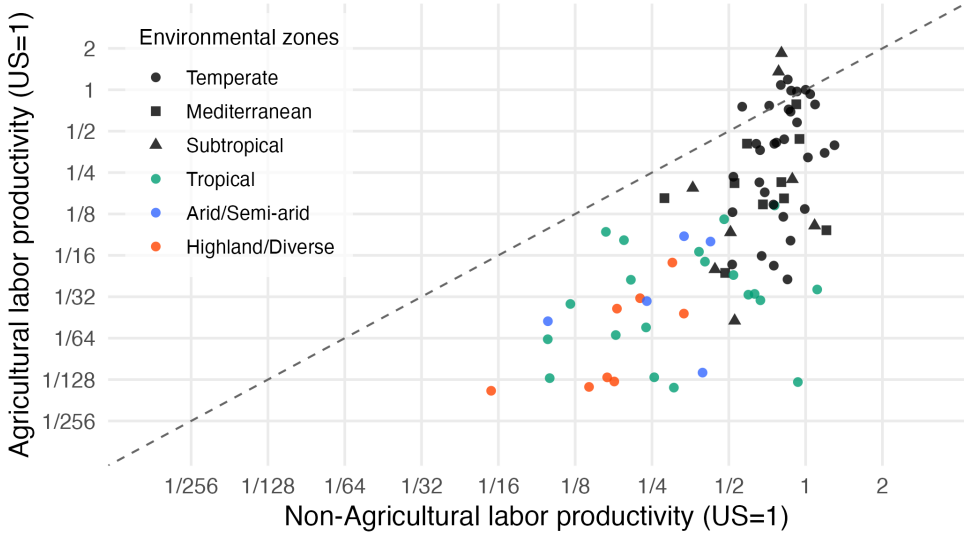
October 27, 2026

Cross-country agricultural productivity differences are enormous ...



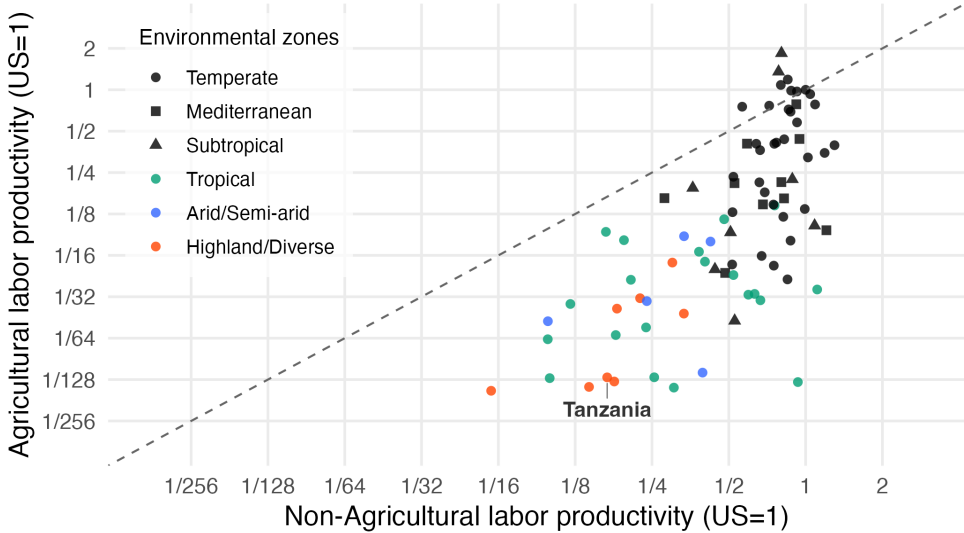
▸ Aggregate ▸ Mfg ▸ Svc ▸ Agri emp share

... particularly where environmental conditions differ from those in US



▸ Map ▸ Classification ▸ Aggregate ▸ Anecdote: maize ▸ Anecdote: cotton

... particularly where environmental conditions differ from those in US



▸ Map ▸ Classification ▸ Aggregate ▸ Anecdote: maize ▸ Anecdote: cotton

How much does directed innovation contribute to cross-country agricultural productivity differences?

Is there role for policy intervention?

This paper

- **Theory:**

- **Quantification:**

- **Counterfactuals:**

This paper

- **Theory:**

  Develops dynamic GE model of appropriate agricultural technology and directed innovation

  Examines directed innovation's role in agricultural productivity gaps and its dynamic efficiency

- **Quantification:**

- **Counterfactuals:**

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  Directed innovation amplifies Tanzania–US agricultural productivity gap by **42%** (1982-2017)

- **Counterfactuals:**

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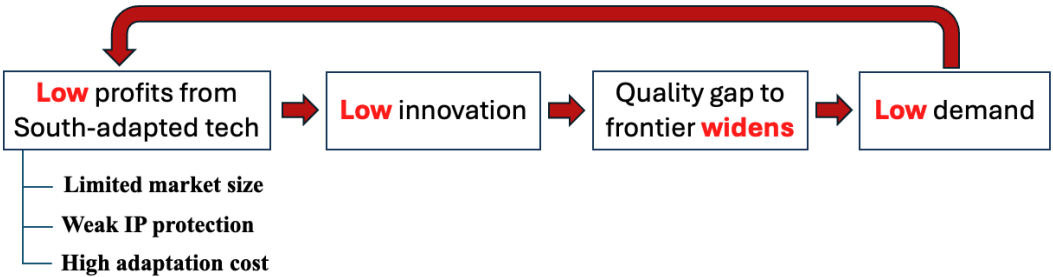
- **Counterfactuals:**

  Locally funded R&D subsidy for Tanzania-adapted technology can narrow gaps & raise welfare

  Stronger IP enforcement in Tanzania has much smaller effects than R&D subsidies

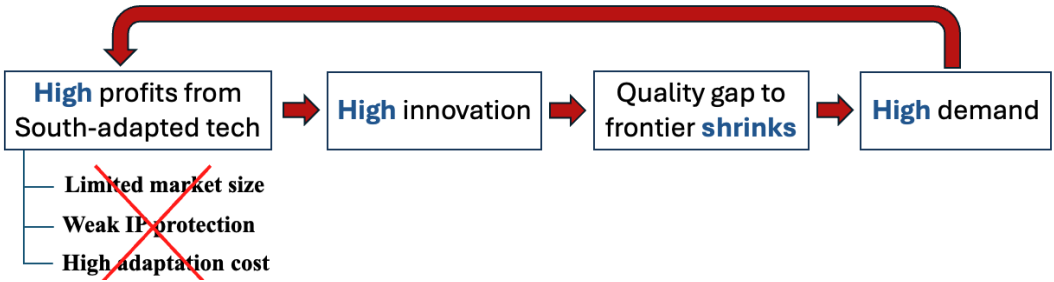
Key mechanism: Dynamic lock-in through directed innovation

- Over time, directed innovation locks Southern farmers into inappropriate North-adapted tech.



Key mechanism: Dynamic lock-in through directed innovation

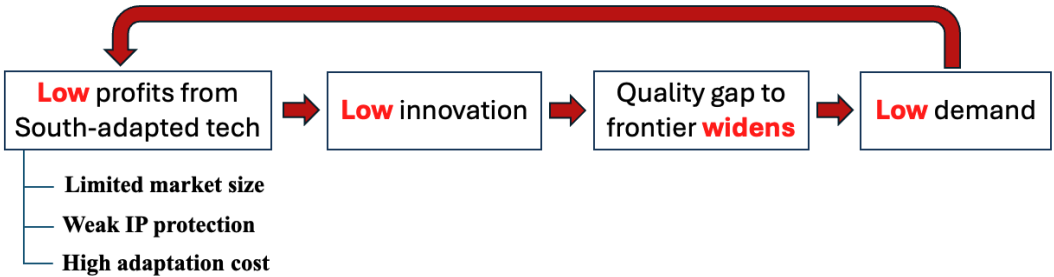
- Dynamic lock-in would not arise if South-adapted technologies were sufficiently profitable



- Innovation in lagging South-adapted tech. incorporates **catch-up via knowledge spillovers**

Key mechanism: Dynamic lock-in through directed innovation

- Over time, directed innovation locks Southern farmers into inappropriate North-adapted tech.



  - Innovation in lagging South-adapted tech. incorporates **catch-up via knowledge spillovers**

- R&D firms do not fully internalize dynamic social gains of innovating in South-adapted tech.

  1. **Lower future agricultural prices**: gains for households
  2. **Better technological base**: gains for future innovators

Key contributions

1. **Cross-country agricultural productivity differences:**

   Caselli (2005), Restuccia-Yang-Zhu (2008), Duflo-Kremer-Robinson (2011), Lagakos-Waugh (2013), Tombe (2015), Gollin-Lagakos-Waugh (2014), Adamopoulos-Restuccia (2014, 2022), Chen (2017, 2020), Gottlieb-Grobovsek (2019), Caunedo-Keller (2021), Donovan (2021), Suri-Udry (2022), Boppart-Kiernan-Krusell-Malmberg (2026)

   → Appropriateness of inputs, shaped by directed innovation, as dynamic source of gaps

2. **Appropriateness of technology:**

   Griliches (1957), Stewart (1978), Basu-Weil (1998), Acemoglu-Zilibotti (2001), Caselli-Coleman (2006), Rossi (2022), Acemoglu-Aghion-Barrage-Hémous (2024), Moscona-Sastry (2025), Akerman-Moscona-Pellegrina-Sastry (2026)

   → Dynamic GE model of lock-in and its inefficiency, built to run policy counterfactuals

3. **Industrial and development policy**

   Harrison-Rodríguez-Clare (2010), Juhász (2018), Itskhoki-Moll (2019), Liu (2019), Juhász-Lane-Rodrik (2024), Choi-Levchenko (2025), Bartelme-Costinot-Donaldson-Rodríguez-Clare (2025), Brooks-Donovan (2026)

   → Dynamic inefficiency in direction of innovation ⇒ self-financed R&D subsidy raises welfare

Outline

Baseline model

Key mechanism & Constrained inefficiency

Quantitative analysis

Counterfactuals

Baseline model

Key mechanism & Constrained inefficiency

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Counterfactuals

Baseline model: Overview

- **Two countries:**
  - *North* and *South* have different environmental conditions
- **Two sectors:**
  - *Agriculture:* Armington differentiation. *Non-agriculture:* homogeneous good, numéraire
- **Agents:**
  1. Perfectly competitive final good producers in each sector
     - Farmers use inputs whose effective quality consists of base quality & mismatch penalty
  2. R&D firms & intermediate input producers
     - Innovators choose which intermediate input type's base quality to improve
  3. Households
     - Endogenous labor allocation: (a) R&D vs. Production, (b) Agriculture vs. Non-agriculture

Agricultural technology in South

$$Y^a_{S,t} = A^a_S (L^a_{S,t})^\beta \left[\left(e^{-\delta} q_{N,t} \right) x_{S,N,t} + q_{S,t} x_{S,S,t} \right]^\alpha \qquad \alpha + \beta < 1, \gamma := 1 - \alpha - \beta$$

- $x_{j,k,t}$: intermediate inputs used in country j and suited to environment k
- $e^{-\delta}$: environmental mismatch penalty
- $q_{k,t}$: base quality of inputs suited to environment k

- Optimal input type:

$$\tau^*_{S,t} = N \quad \text{if} \quad \frac{e^{-\delta} q_{N,t}}{p^x_{N,t}} > \frac{q_{S,t}}{p^x_{S,t}}; \quad S \text{ otherwise}$$

		Input type N	Input type S
Country	S	$e^{-\delta} q_{N,t}$	$q_{S,t}$
	N	$q_{N,t}$	$e^{-\delta} q_{S,t}$

- Optimal labor demand:

$$w^a_{S,t} = \beta \frac{p^a_{S,t} Y^a_{S,t}}{L^a_{S,t}}$$

Non-agricultural technology in South

$$Y^n_{S,t} = A^n_S L^n_{S,t}$$

- Optimal labor demand:

$$w^n_{S,t} = A^n_S$$

- No-arbitrage condition:

$$w^n_{S,t} = w^a_{S,t} \quad \rightarrow \quad L^a_{S,t} = \beta \frac{p^a_{S,t} Y^a_{S,t}}{A^n_S}$$

R&D firms/intermediate input producers: Timing

1. **Entry + R&D:**
   - Potential R&D firms choose whether to innovate & which input type to improve

2. **Production:**
   - Intermediate input producer sets prices as monopolist & sells to all countries

R&D firms/intermediate input producers: Timing

1. **Entry + R&D:**
   - Potential R&D firms choose whether to innovate & which input type to improve

2. **Production:**
   - Intermediate input producer sets prices as monopolist & sells to all countries
   - Profit function of type-k monopolist in period t:

$$\pi_{k,t} := \underbrace{\left(1 - \frac{q_{k,t-1}}{q_{k,t}}\right)}_{\text{margin rate}} \underbrace{\sum_{\{j:\tau^*_{j,t}=k\}} \alpha p^a_{j,t} Y^a_{j,t}}_{\text{market size}}, \qquad \tau^*_{j,t}\text{: input type used in country } j$$

▸ Stage-2 problem

R&D firms/intermediate input producers: Timing

1. **Entry + R&D:**
   - Potential R&D firms choose whether to innovate & which input type to improve
   - Firm pays R&D costs to improve type-k inputs, receiving one-period patent
   - At most one entrant per type per period, ensured by sequential entry & patent exclusion
   - R&D occurs only in North

▸ Endogenous R&D location

2. **Production:**
   - Intermediate input producer sets prices as monopolist & sells to all countries
   - Profit function of type-k monopolist in period t:

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▸ Stage-2 problem

Stage 1: Entry and R&D decisions

- Quality ladder:
  - Northern inputs: $q_{N,t} = (1+\xi) q_{N,t-1}$ if innovated at time t; $q_{N,t} = q_{N,t-1}$ otherwise
  - Southern inputs:

$$q_{S,t} = \begin{cases} (1+\xi)\left[a \cdot \chi \cdot q_{N,t-1} + (1-a) \cdot q_{S,t-1}\right] & \text{if innovated at time } t, \\ q_{S,t-1} & \text{otherwise.} \end{cases}$$

    - $a \in (0,1]$: adaptation of frontier technology; $\chi \in (0,1]$: transferability of frontier technology
    - ***Advantage of backwardness (Gerschenkron, 1962)***

- R&D cost for any input type:

$$c := w_N^r / \eta$$

  - Improving tyke-k's base quality requires $1/\eta$ scientists at wage w_N^r.

- Directed innovation:
  1. **Only Northern inputs are innovated:** $\pi_{N,t} \geq c$, $\pi_{S,t} < c$
  2. Both input types are innovated: $\pi_{N,t} \geq c$, $\pi_{S,t} \geq c$

Agricultural goods are imperfect substitutes across origins (Armington)

- **Agricultural consumption in South:**

$$C^a_{S,t} = \left[\mu^{\frac{1}{\varepsilon}}(c^a_{S,N,t})^{\frac{\varepsilon-1}{\varepsilon}} + (1-\mu)^{\frac{1}{\varepsilon}}(c^a_{S,S,t})^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}}, \qquad \varepsilon > 1$$

  - $c_{j,k,t}$: country j's consumption of agricultural goods produced in country k

- Aggregate consumption in South:

$$C_{S,t} = (C^a_{S,t})^{\phi}(C^n_{S,t})^{1-\phi}$$

▸ Households ▸ RC

Outline

Baseline model

Key mechanism & Constrained inefficiency

Quantitative analysis

Counterfactuals

Agricultural labor productivity differences between South and North

$$\frac{Y^a_{S,t}/L^a_{S,t}}{Y^a_{N,t}/L^a_{N,t}} = \left[\left(\frac{A^a_S}{A^a_N}\right)\cdot\left(\frac{A^n_S}{A^n_N}\right)^{\alpha+\gamma\varepsilon}\cdot\mathcal{Q}(\lambda_{t-1})^{\alpha}\right]^{\frac{1}{1+\gamma(\varepsilon-1)}},\quad \mathcal{Q}(\lambda) := \max\left\{\lambda, e^{-\delta}\right\},\quad \lambda := \frac{q_S}{q_N}$$

▸ Quantitative

1. Agricultural fundamentals
2. Non-agricultural productivity
3. Directed innovation & Environmental mismatch

Agricultural labor productivity differences between South and North

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▸ Quantitative

1. Agricultural fundamentals
2. Non-agricultural productivity
3. **Directed innovation** & **Environmental mismatch**

- Two scenarios:
  (a) Dynamic lock-in through directed innovation
  (b) No dynamic lock-in

Dynamic lock-in widens agricultural productivity differences over time

Directed innovation

State of quality ratio $\rightarrow$ Profitability $\rightarrow$ Innovation outcome $\rightarrow$ Resulting quality ratio

$\lambda_{t-1} = \frac{q_{S,t-1}}{q_{N,t-1}}$ $\quad \pi_{S,t}(\lambda_{t-1}) < c$ No innovation in S-inputs $\quad \lambda_t < \lambda_{t-1}$

Small market size

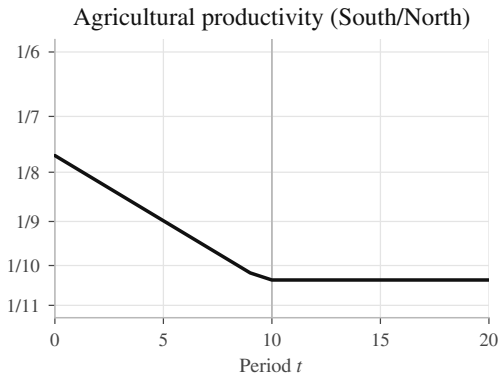
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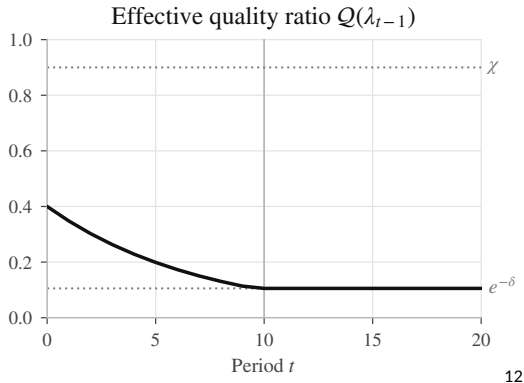
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Directed innovation





Undirected innovation benchmark: No lock-in & catch-up

Undirected innovation

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$\lambda_{t-1} = \frac{q_{S,t-1}}{q_{N,t-1}}$ Innovation in both input types $\lambda_t > \lambda_{t-1}$

Undirected innovation benchmark: No lock-in & catch-up

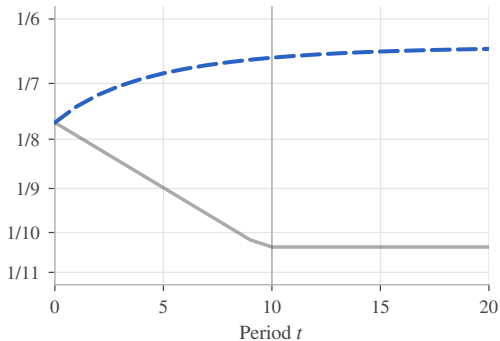
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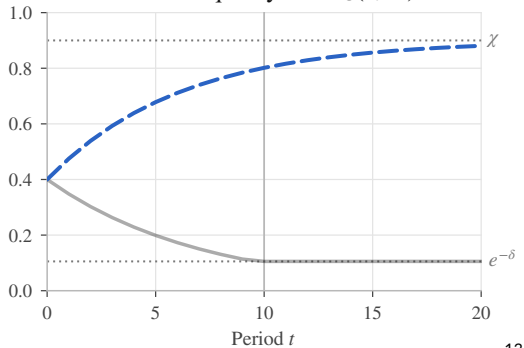
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Directed innovation — Undirected innovation

Agricultural productivity (South/North)



Effective quality ratio $\mathcal{Q}(\lambda_{t-1})$



Dynamic inefficiency & Constrained social planner

- To isolate *dynamic* inefficiency from *static* monopoly distortion, consider planner subject to

  1. same technological and feasibility constraints as in decentralized economy;
  2. **structure of within-period decentralized equilibrium, including monopoly distortion**

- Planner chooses whether to improve S-inputs, ***on behalf of R&D firms***, to max Southern welfare

▸ Equation

Wedge between private and social innovation incentives

- Private agents do not fully internalize **continuation value** of undertaking type-S innovation:

  1. **Lower future food prices**: gains for households

  2. **Better technological base**: gains for future innovators

- **Missing intertemporal market**: original innovator has no tradable claim on these future gains

▸ Equation

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     S-input quality $\uparrow \rightarrow$ Southern agricultural productivity $\uparrow \rightarrow p^a_{S,t} \downarrow \rightarrow$ Agricultural price index $\downarrow$

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  2. **Better technological base**: gains for future innovators

     S-input quality $\uparrow \rightarrow$ larger S-market (including through earlier adoption) $\rightarrow$ profitability $\uparrow$

- **Missing intertemporal market**: original innovator has no tradable claim on these future gains

▸ Equation

Outline

Baseline model

Key mechanism & Constrained inefficiency

Quantitative analysis

Counterfactuals

Quantitative model: Seven differences from baseline model

- Relaxing restrictive assumptions in baseline model:

1. R&D opportunities on every product line $\ell \in [0, 1]$ within input type $k \in \{N, S\}$ ▸ R&D

2. Continuum of farmers $i \in [0, 1]$ in each country j & Fréchet shocks to farmers ▸ Fréchet

3. Weak IP enforcement and imitators in South & Higher fixed costs for type-S innovation ▸ IP & FC

- Adding features to better match data:

4. Sectoral labor wedges; 5. Stone-Geary preferences; 6. Iceberg trade costs; 7. Land ▸ Additional ▸ APD

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   $\rightarrow$ ***Continuous* type-k innovation intensity:** $m_{k,t} \in [0, 1]$

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   $\rightarrow$ ***Continuous* type-k adoption share in country j:** $\omega_{j,k,t} \in [0, 1]$

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3. Weak IP enforcement and imitators in South & Higher fixed costs for type-S innovation ▸ IP & FC

   → **Captures institutional barriers $\zeta_S < 1$ & higher R&D costs for type-S innovators**

- Adding features to better match data:

4. Sectoral labor wedges; 5. Stone-Geary preferences; 6. Iceberg trade costs; 7. Land ▸ Additional ▸ APD

Calibration: Externally calibrated parameters

- One period = 5 years. Calibration targets: 2017 moments. **South** = Tanzania. **North** = US

Parameter	Description	Value	Source
α, β, γ	Factor shares (inputs, labor, land)	0.40, 0.42, 0.18	Restuccia-Yang-Zhu (2008); Donovan (2021)
θ	Fréchet parameter	2.46	Costinot-Donaldson-Smith (2016)
ε	Agricultural Armington elasticity	5.4	Costinot-Donaldson-Smith (2016)
ξ	Frontier quality step size	0.177	US agricultural TFP growth (1982–2017)
ρ, σ	Discount rate, Utility curvature	0.0515, 1	1% per year, Standard
δ	Environmental mismatch penalty	1.45	Moscona-Sastry (2025)
A_S^n / A_N^n	Non-agricultural productivity ratio	0.167	Non-agricultural VA per worker (2017)
Z_S / Z_N	Land endowment ratio	0.093	Cropland (2017)
L_S / L_N	Labor endowment ratio	0.130	Persons engaged (2017)
ζ_S	Enforcement of IP rights	0.78	Campi-Nuvolari (2021)
τ_g	Trade cost on agricultural goods	4.58	Own-supply share of food (2017)
$1+\tau_{w,N}$, $1+\tau_{w,S}$	Labor wedges	1.75, 21.9	Sectoral gaps in VA per worker (2017)
f	Additional type-S R&D fixed cost	0.34	Phillips McDougall (2011, 2016); AgbioInvestor (2024, 2026)

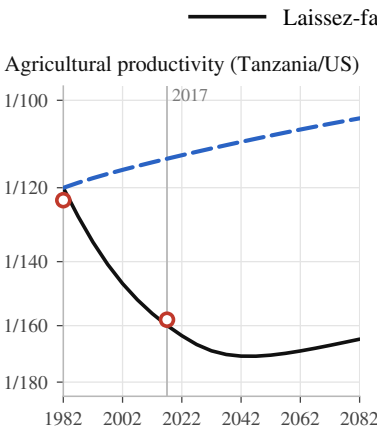
▸ MS ▸ IP

Calibration: Internally calibrated parameters

Parameter	Description	Value	Target moment	Data	Model
ϕ	Agricultural expenditure share	0.0137	US agricultural employment share (2017)	0.0137	0.0137
η	Research productivity	4338	US private agricultural-input R&D intensity (2010–14)	0.0124	0.0124
A_S^a	Southern agricultural fundamentals ($A_N^a = 1$)	0.118	Agricultural productivity differences (2017)	1/160	1/160
$\bar{a}$	Subsistence consumption per capita	0.0165	East Africa's share of farm output (2017)	0.156	0.156
b	R&D fixed-cost distribution half-width	0.73	Private breeder presence in East Africa (TASAI, 2021)		
a, χ	Adaptation rate, Transferability	0.29, 0.65	Modern variety adoption during the Green Revolution in Evenson-Gollin (2003)		
λ_{2017}	Quality ratio inherited in 2017	0.236	Modern variety shares and ages of varieties in use		

▸ Model fit ▸ GR ▸ Initial state

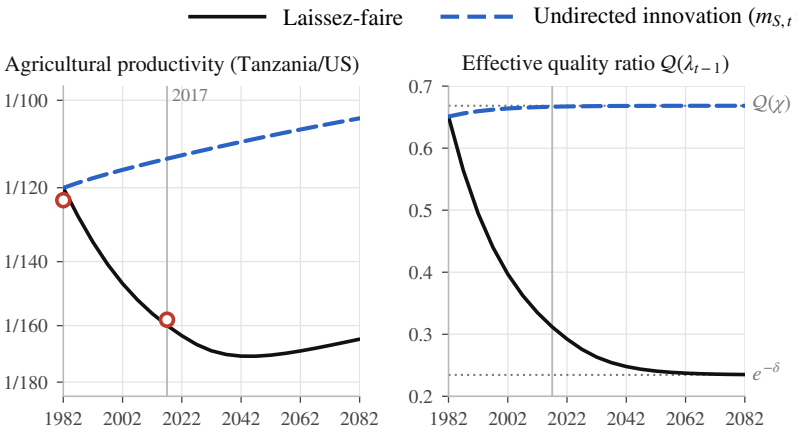
Directed innovation contributes to productivity gaps by 42% in 2017



- Relative to undirected innovation counterfactual: **42%** in 2017, **57%** in 2047, **51%** in long run
- Two channels: (1) **effective quality of inputs** & (2) **structural change**

▸ Longer horizon

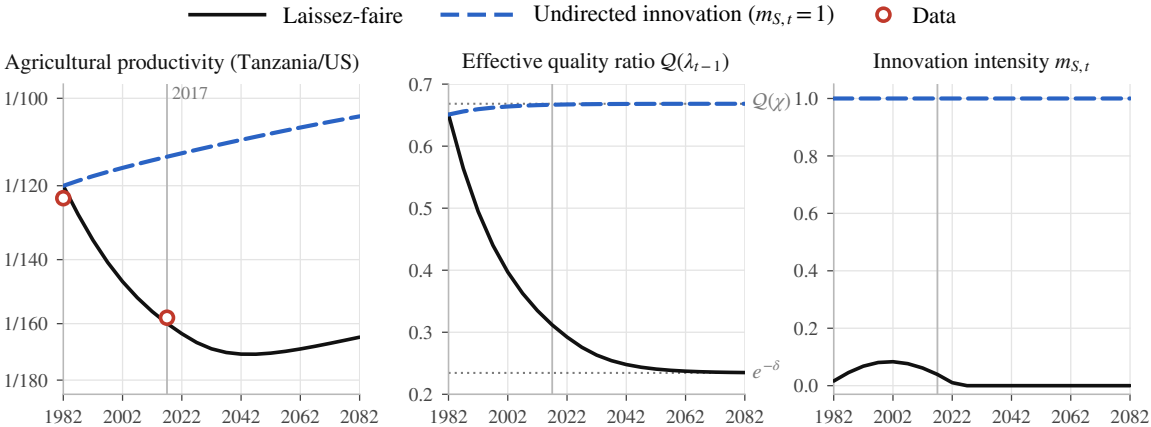
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Counterfactuals

Three counterfactuals: IP reform, small and large R&D subsidies

1. Perfect IP enforcement in South: $\zeta_S = 1$

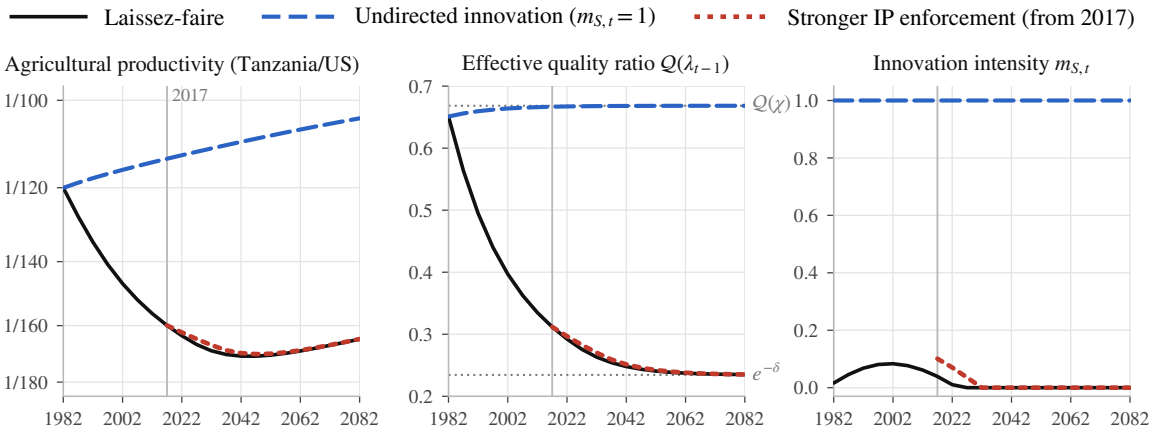
2. Small R&D subsidy: $m_{S,t}$ such that $\lambda_t = \lambda_{t-1}$

3. Large R&D subsidy: $m_{S,t} = 1$

   - Both subsidies are financed by lump-sum tax on Southern households

   - Policies are implemented from 2017 onward, with model starting in 1982

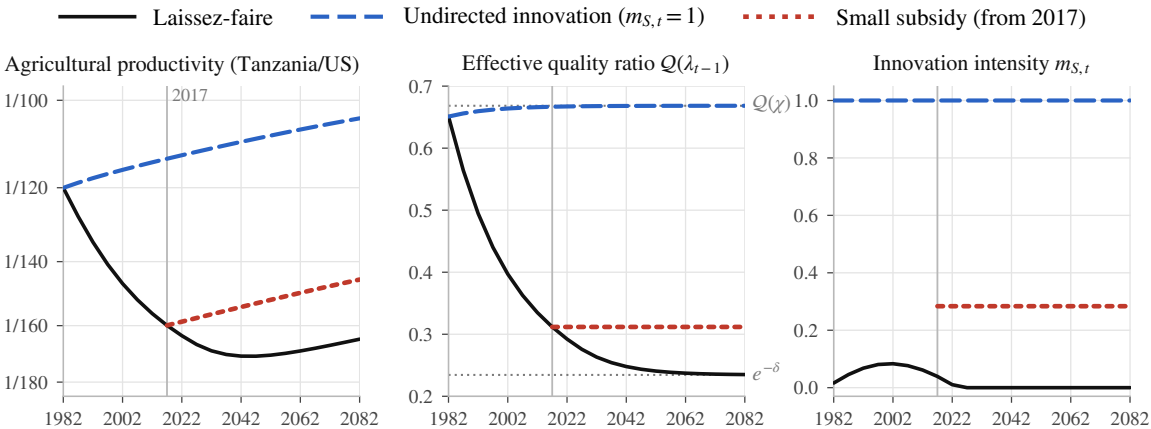
IP reform does little to narrow agricultural productivity differences



- Tanzania's agri productivity relative to US rises by **0.9%** in 2037, **0.4%** in 2047, **0%** in long run
- Welfare gains for Southern households: **0.1%**

▸ Longer horizon ▸ welfare

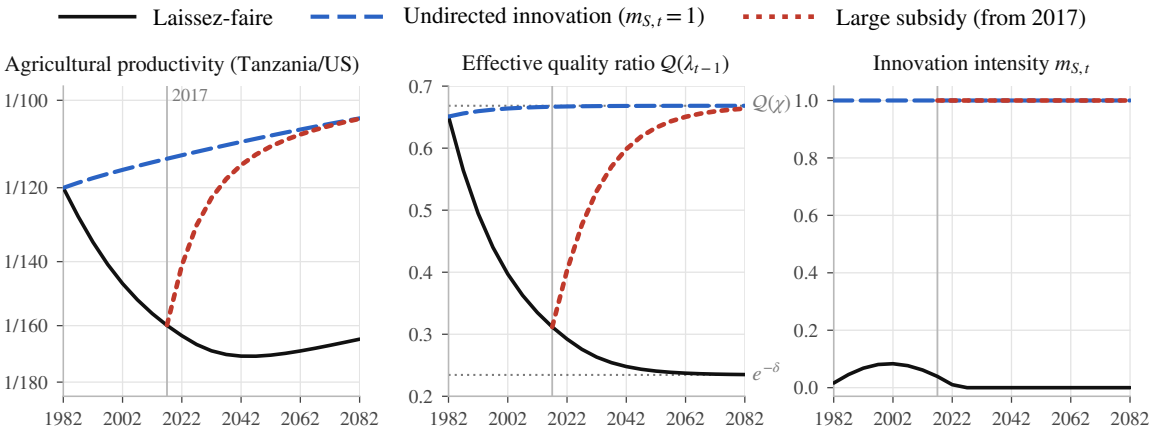
Small R&D subsidy narrows productivity differences & raises welfare



- Tanzania's agri productivity relative to US rises by **9.6%** in 2037, **11.6%** in 2047, **12%** in long run
- Welfare gains for Southern households: **2.4%**

▸ Subsidy cost ▸ Longer horizon

Large R&D subsidy substantially narrows productivity differences



- Tanzania's agri productivity relative to US rises by **44%** in 2037, **52%** in 2047, **51%** in long run
- Welfare gains for Southern households: **6.76%**

▸ Subsidy cost ▸ Longer horizon

Small R&D subsidy raises welfare under both wage gap interpretations

	EV_S (%)	Food price index	Subsistence cost	Labor income	Land income	Imitators' profits	Tax
Small R&D subsidy							
Barrier interpretation	**2.40**	0.13	0.37	2.24	−0.05	0.01	−0.30

- Large gains from labor reallocation from agriculture to non-agriculture

▸ welfare

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▸ welfare

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Effective-labor interpretation	**0.10**	0.13	0.53	0.00	−0.08	0.00	−0.49

- Large gains from labor reallocation from agriculture to non-agriculture
- Two views on sectoral wage gap: (1) friction view vs. (2) frictionless view
  - Regardless of interpretations, small R&D subsidy generates welfare gains

▸ welfare

Small R&D subsidy raises welfare under both wage gap interpretations

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Small R&D subsidy							
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Effective-labor interpretation	**0.10**	0.13	0.53	0.00	−0.08	0.00	−0.49
Large R&D subsidy							
Barrier interpretation	**6.76**	0.52	1.26	8.57	−0.18	0.03	−3.44
Effective-labor interpretation	**−3.23**	0.50	1.82	0.00	−0.29	0.02	−5.28

- Large gains from labor reallocation from agriculture to non-agriculture
- Two views on sectoral wage gap: (1) friction view vs. (2) frictionless view
  - Regardless of interpretations, small R&D subsidy generates welfare gains
  - Welfare effects of large R&D subsidy depend on wage gap interpretation

▸ welfare

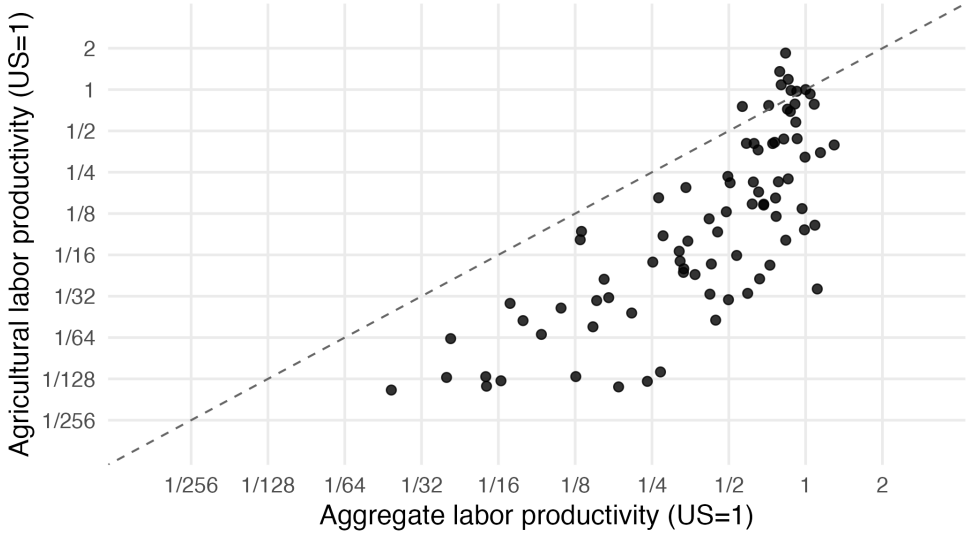
Conclusion

- Directed innovation contributes to Tanzania–US agricultural productivity gap by **42%**
- R&D policies that redirect innovation can narrow these gaps and raise welfare
- Extension: Environmental heterogeneity & Innovator's tailoring decisions ▸ Env hetero
- **Research agenda:**

  Economic structure across stages of development: consequences for growth & policy

  1. Cross-Country Differences in the Impacts of Sectoral Shocks
  2. Informality and the Job Ladder in Developing Countries, with *Matthias Doepke*
  3. The Size-Centrality Relationship in Production Networks, with *Nikola Dacic* & *Marko Melolinna*
  4. Macroprudential Regulation, Financial Stability, and Economic Growth

Agricultural productivity diff is larger than Aggregate prod diff in 2017

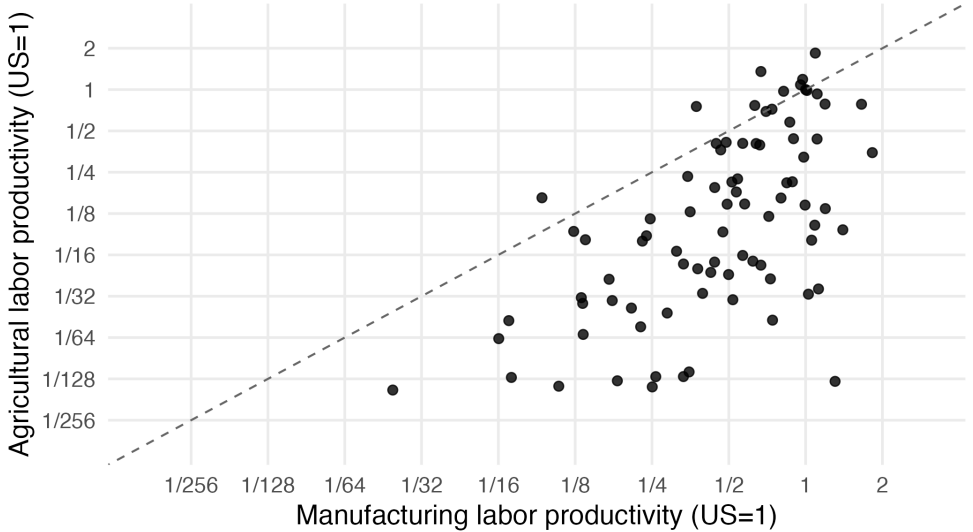


Source: GGDC Productivity Level Database 2023

x-axis: PPP GDP per worker, *y*-axis: PPP Agricultural value added per worker

Back

Agricultural productivity diff is larger than Manufacturing prod diff ’17

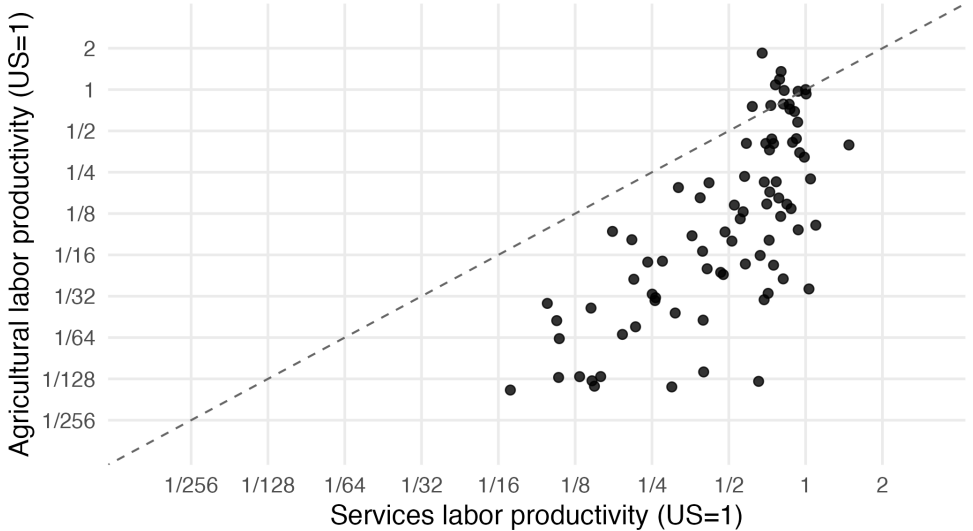


Source: GGDC Productivity Level Database 2023

x-axis: PPP Manufacturing value added per worker, *y*-axis: PPP Agricultural value added per worker

Back

Agricultural productivity diff is larger than Services prod diff ’17

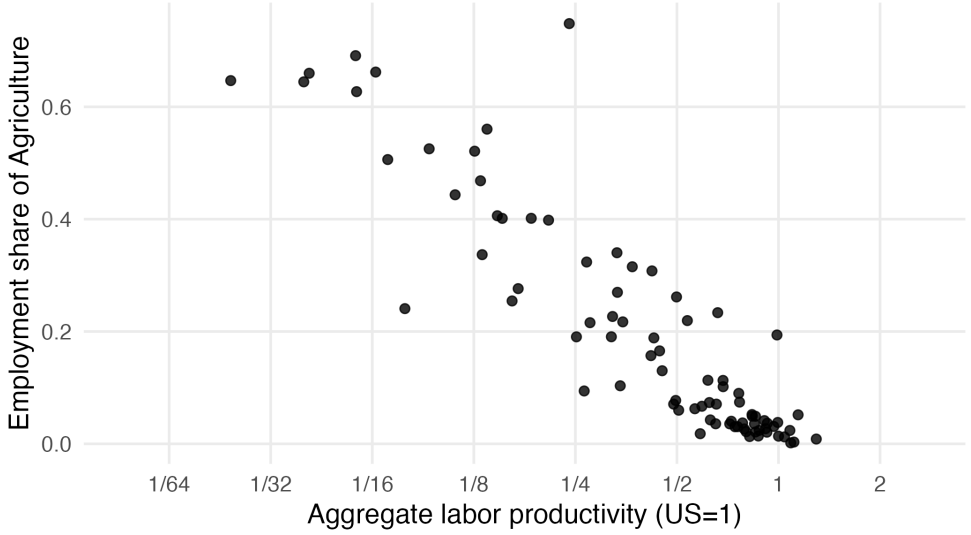


Source: GGDC Productivity Level Database 2023

x-axis: PPP Services value added per worker, *y*-axis: PPP Agricultural value added per worker

▸ Back

Employment share of Agri decreasing in Aggregate productivity in ’17

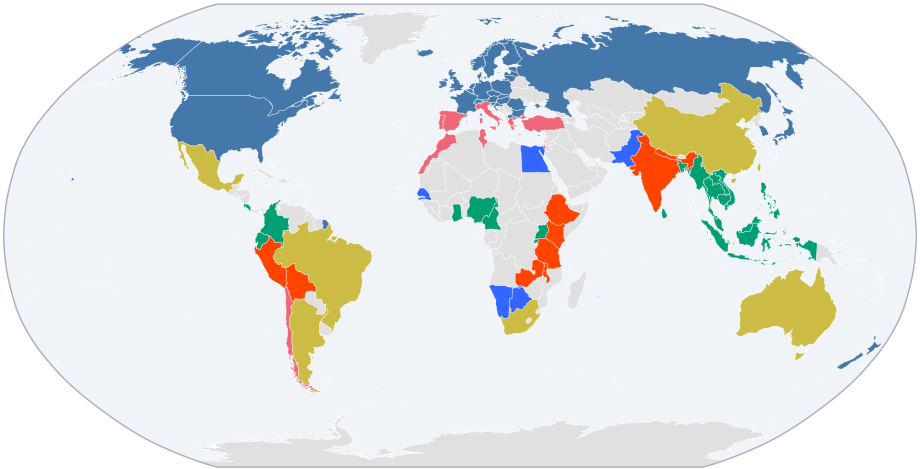


Source: GGDC Productivity Level Database 2023

x-axis: PPP GDP per worker

Back

Environmental zones: world map



Agro-climatic zone Temperate Mediterranean Subtropical Tropical Arid/Semi-arid Highland/Diverse NA

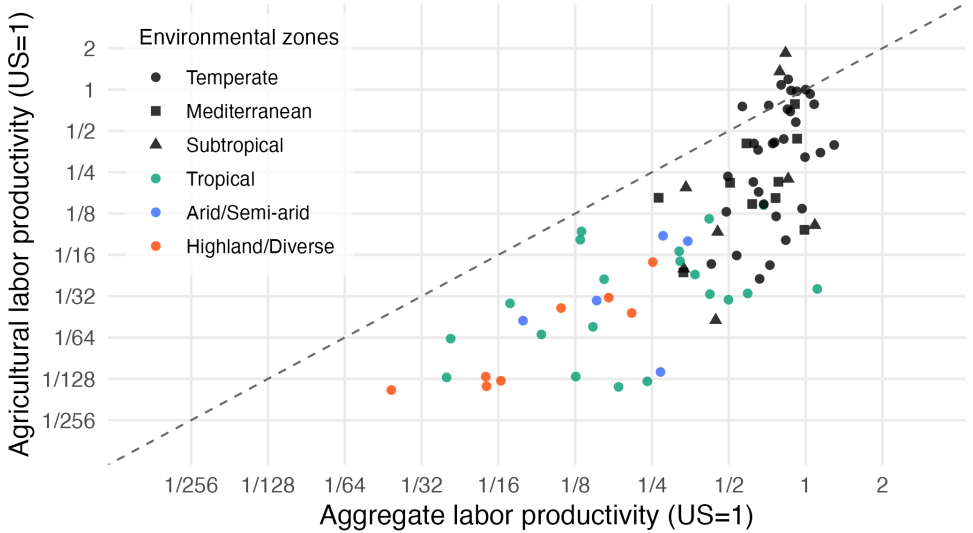
Back

Environmental zones: classification

1. **Temperate:**
   - Countries whose farmland lies in Köppen C (oceanic/humid) or D (continental) zones
   - 32 countries: Most of Europe, Canada, Japan, Korea, Russia, New Zealand
2. **Mediterranean:**
   - Dry-summer subtropical climates (Csa/Csb); similar to California agriculture
   - 10 countries: Southern Europe, Turkey, the Maghreb, Chile, Israel
3. **Subtropical humid:**
   - Warm humid zones (Cfa/Cwa/Cwb); between temperate and tropical, similar to the US Southeast.
   - 8 countries: Argentina, Australia, Brazil, China, Taiwan, Mexico, South Africa, Hong Kong
4. **Tropical humid:**
   - Year-round warm with high rainfall (Af/Am/Aw); very different from US agriculture.
   - 20 countries: Southeast Asia, West/Central Africa, Central America, Bangladesh, Sri Lanka
5. **Arid/Semi-arid:**
   - Moisture-limited agriculture (BWh/BSh): 5 countries: Egypt, Pakistan, Botswana, Namibia, Senegal
6. **Highland/Diverse:**
   - Countries with highly heterogeneous agro-climatic conditions.
   - 9 countries: Ethiopia, India, Kenya, Nepal, Bolivia, Peru, Tanzania, Malawi, Zambia

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Agricultural productivity diff is larger than Aggregate prod diff in 2017



Source: GGDC Productivity Level Database 2023

x-axis: PPP GDP per worker, *y*-axis: PPP Agricultural value added per worker

Back

Maize resistant to African stem borers was not available to Kenyan farmers in 2006



- Maize was destroyed by stem borers, an African pest, in Embu, Kenya, 2006

- The resistant maize that existed had been developed for North American borers

Source: CIMMYT (2006)

Back

Introduction of Monsanto's genetically modified cotton in Burkina Faso



- In 2008, US-bred insect-resistant (Bt) trait was crossed into Burkina Faso's long-staple varieties
- Quality of Burkinabé cotton then declined
- Bt trait had been optimized for US pest (bollworm)
- Careful local adaptation was needed

Source: Dowd-Uribe-Schnurr (2016), Reuters (2017)

Back

Endogenous R&D location

- R&D cost for any input type:

$$c_j := \frac{w_j^n}{\eta_j}$$

  - $\eta_N \gg \eta_S$: research productivity. Creating new vintage requires $1/\eta_j$ scientists at wage w_j^n

- Net profit of R&D firm located in country j conducting R&D for type-k inputs:

$$\pi_{k,t} - c_j$$

- **Lemma:** R&D occurs in North but not in South when the following condition holds:

$$c_N < \pi_{N,t} < c_S \quad \Leftrightarrow \quad \frac{A_N^n}{\eta_N} < \pi_{N,t} < \frac{A_S^n}{\eta_S}$$

  - In South, research productivity is so low that R&D is never profitable

Back

Stage 2: Production

- One unit of each input type is produced at MC $\psi = \alpha^2$ (in units of non-agri goods)

- Type-k input monopolist chooses price $p^x_{k,t}$ to maximize

$$\sum_{\{j:\tau^*_{j,t}=k\}} \left[p^x_{k,t} - \psi\right] x_{j,k,t}(p^x_{k,t}), \qquad \tau^*_{j,t}\text{: input type used in country } j$$

- Limit pricing:

$$p^x_{k,t} = \frac{q_{k,t}}{q_{k,t-1}}\psi$$

- Profit function:

$$\pi_{k,t} := \underbrace{\left(1 - \frac{q_{k,t-1}}{q_{k,t}}\right)}_{\text{margin rate}} \underbrace{\left[\mathbb{1}\{\tau^*_{N,t} = k\}\alpha p^a_{N,t} Y^a_{N,t} + \mathbb{1}\{\tau^*_{S,t} = k\}\alpha p^a_{S,t} Y^a_{S,t}\right]}_{\text{market size}}$$

Back

Households

- Preferences:

$$U_j = \sum_{t=0}^{\infty} \frac{1}{(1+\rho)^t} u(C_{j,t})$$

where

$$C_{j,t} = (C^a_{j,t})^{\phi} (C^n_{j,t})^{1-\phi}, \quad C^a_{j,t} = \left[\mu^{\frac{1}{\varepsilon}} (c^a_{j,N,t})^{\frac{\varepsilon-1}{\varepsilon}} + (1-\mu)^{\frac{1}{\varepsilon}} (c^a_{j,S,t})^{\frac{\varepsilon-1}{\varepsilon}} \right]^{\frac{\varepsilon}{\varepsilon-1}}$$

- Period budget constraints:

$$p^a_{N,t} c^a_{j,N,t} + p^a_{S,t} c^a_{j,S,t} + C^n_{j,t} = E_{j,t},$$

where

$$E_{N,t} = w^a_{N,t} L^a_{N,t} + w^n_{N,t} L^n_{N,t} + w^r_{N,t} L^r_{N,t} + \text{Profit}^a_{N,t} + \sum_{k \in \{N,S\}} \Pi_{k,t},$$

$$E_{S,t} = w^a_{S,t} L^a_{S,t} + w^n_{S,t} L^n_{S,t} + \text{Profit}^a_{S,t},$$

- $\text{Profit}^a_{j,t}$: profits of agricultural final good producers in country j
- $\Pi_{k,t} := \pi_{k,t} - c$: net profits of type-$k$ R&D firms

Back

Optimality conditions of households

- Agricultural varieties produced in North vs. South:

$$c^a_{j,k,t} = \mu \left(\frac{p^a_{k,t}}{P^a_t} \right)^{-\varepsilon} C^a_{j,t}, \qquad P^a_t = \left[\mu (p^a_{N,t})^{1-\varepsilon} + (1-\mu)(p^a_{S,t})^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}$$

- Agricultural vs. non-agricultural goods:

$$P^a_t C^a_{j,t} = \omega E_{j,t}, \qquad C^n_{j,t} = (1-\omega) E_{j,t}$$

where

$$P^a_t C^a_{j,t} + C^n_{j,t} = E_{j,t}$$

Back

Market-clearing conditions

- Agricultural goods:

$$c^a_{N,j,t} + c^a_{S,j,t} = Y^a_{j,t}$$

- Non-agricultural goods:

$$C^n_{N,t} + C^n_{S,t} + \sum_{k \in \{S,N\}} \psi(x_{N,k,t} + x_{S,k,t}) = Y^n_{N,t} + Y^n_{S,t}$$

- Labor:

$$\underbrace{L^a_{N,t} + L^n_{N,t}}_{\text{workers}} + \underbrace{L^r_{N,t}}_{\text{researchers}} = L_N \quad \text{and} \quad L^a_{S,t} + L^n_{S,t} = L_S$$

Back

Dynamic inefficiency & Constrained social planner's problem

- Planner chooses whether to improve S-inputs ($m_{S,t} \in \{0,1\}$), ***on behalf of R&D firms***, to max

$$\sum_{t=0}^{\infty} \frac{1}{(1+\rho)^t} \left[v_N u(C_{N,t}) + v_S u(C_{S,t}) \right], \quad v_j \geq 0\text{: Pareto weights}$$

  - (1) S-planner $v_S = 1, v_N = 0$; **(2) Global planner** $v_S, v_N > 0$; (3) N-planner $v_S = 0, v_N = 1$

- Bellman equation:

$$V^{SP}(\lambda) = \max_{m_S \in \{0,1\}} \left\{ \widetilde{w}(\lambda, m_S) + \frac{1}{1+\rho} V^{SP}\left(\lambda'(\lambda, m_S)\right) \right\},$$

s.t. $\widetilde{w}(\lambda, m_S) = \left[v_N u(c_N) + v_S u(c_S) \right](\lambda, m_S)$, $\lambda'(\lambda, m_S) = \begin{cases} a\chi + (1-a)\lambda & \text{if } m_S = 1, \\ \lambda/(1+\gamma) & \text{if } m_S = 0. \end{cases}$

Back

Wedge between private and social innovation incentives

- Social and private values of undertaking type-S innovation:

$$\Pi^{SP}(\lambda) = \underbrace{\left[\widetilde{w}(\lambda, 1) - \widetilde{w}(\lambda, 0)\right]}_{\text{instantaneous payoff}} + \underbrace{\frac{1}{1+\rho}\left[V^{SP}\left(\lambda'(\lambda, 1)\right) - V^{SP}\left(\lambda'(\lambda, 0)\right)\right]}_{\text{continuation value}}$$

$$\Pi(\lambda) = \underbrace{\pi_S(\lambda) - c_N}_{\text{instantaneous profit}}$$

- Continuation value is typically large enough that $\Pi^{SP}(\lambda) \geq 0$, while $\Pi(\lambda) < 0$

Back

Directed innovation with intensive margin: hetero. R&D costs across ℓ

- Firm innovating type-k inputs has research opportunities on all product lines $\ell \in [0,1]$
- Manager of line ℓ observes its R&D fixed cost $c_{k,\ell,t}$ and decides whether to improve it
  - $c_{k,\ell,t}$ is stochastic & independent across ℓ, k, t with cdf $H(c/c_k)$, $\mathbb{E}[c_{k,\ell,t}] = c_k$, and

$$c_N = w_N^r/\eta, \quad c_S = (1+f)\, c_N$$

  - Improves line ℓ iff $\pi_{k,\ell,t} \geq c_{k,\ell,t}$. In eqm, $\pi_{k,\ell,t}$ will be same for all ℓ, so write $\pi_{k,t} := \pi_{k,\ell,t}$
- **Type-k innovation *intensity*:** $m_{k,t} := \int_0^1 \mathbb{1}\{\pi_{k,t} \geq c_{k,\ell,t}\}\, \mathrm{d}\ell = H(\pi_{k,t}/c_k) \in [0,1]$
- Law of motion:

$$\Delta \log \lambda_t = \Delta \log q_{S,t} - \Delta \log q_{N,t},$$

where

$$\Delta \log q_{S,t} = m_{S,t} \log\left[\frac{(1+\xi)(a\chi + (1-a)\lambda_{t-1})}{\lambda_{t-1}}\right] \quad \text{and} \quad \Delta \log q_{N,t} = m_{N,t} \log(1+\xi).$$

  - Quality of input type k: $\log q_{k,t} = \int_0^1 \log q_{k,\ell,t} \mathrm{d}\ell$

▸ Back

Innovation decision for product line ℓ of input type k

- Profit function for line ℓ of type k (from stage 2):

$$\pi_{k,\ell,t} := \underbrace{\left(1 - \frac{q_{k,t-1}}{q_{k,\ell,t}}\right)}_{\text{margin rate}} \underbrace{\left[\omega_{N,k,t} \cdot \alpha p^a_{N,t} Y^a_{N,t} + \zeta_S \cdot \omega_{S,k,t} \cdot \alpha p^a_{S,t} Y^a_{S,t}\right]}_{\text{market size}}$$

  - $\omega_{j,k,t}$: share of farmers in country j using type-k inputs
  - **Weak IP enforcement:** monopolist can appropriate fraction $\zeta_S \leq 1$ of profits from South
    - Fraction $(1 - \zeta_S)$ of profits accrues to Southern imitators (owned by Southern households)
- **Differential R&D costs** for type-k innovation: $c_N = w^r_N / \eta_N$ and $c_S = c_N(1 + f)$
  - Northern firms have less local agronomic knowledge in South, requiring more field trials
- Quality ladder:

$$q_{N,\ell,t} = (1 + \gamma) q_{N,t-1}, \quad q_{S,\ell,t} = (1 + \tilde{\xi}) \left[a \cdot \chi \cdot q_{N,t-1} + (1 - a) \cdot q_{S,t-1}\right] \quad \text{if improved}$$

  - Innovation builds on *type*-level public knowledge $q_{k,t-1}$

Back

Input type choice with intensive margin: Fréchet shocks to farmers

- Agricultural production function of farmer $i \in [0, 1]$ in South:

$$Y^a_{S,t}(i) = A^a_S L^a_{S,t}(i)^\beta \underbrace{Z^\gamma_S}_{\text{land}} \int_0^1 \left[\left(e_{S,N,t}(i) e^{-\delta} q_{N,\ell,t} \right) x_{S,N,\ell,t}(i) + \left(e_{S,S,t}(i) \cdot q_{S,\ell,t} \right) x_{S,S,\ell,t}(i) \right]^\alpha \mathrm{d}\ell$$

  - $e_{j,k,t}(i)$ is drawn from Fréchet dist, iid across i, k, t: $P(e_{j,k,t}(i) \leq b) = \exp\{-b^{-\theta}\}$

- Fraction of Southern farmers using Northern and Southern inputs:

$$\omega_{S,N,t} = \frac{(e^{-\delta} q_{N,t-1})^\theta}{(e^{-\delta} q_{N,t-1})^\theta + (q_{S,t-1})^\theta}, \qquad \omega_{S,S,t} = \frac{(q_{S,t-1})^\theta}{(e^{-\delta} q_{N,t-1})^\theta + (q_{S,t-1})^\theta}$$

- Derived country-level agricultural production function:

$$Y^a_{S,t} = A^a_S (L^a_{S,t})^\beta Z^\gamma_S \left[\mathcal{Q}_{S,t} X_{S,t} \right]^\alpha \cdot \Gamma_\theta$$

▸ Back

  - $X_{S,t} (:= x_{S,N,t} + x_{S,S,t})$: aggregate intermediate input. $\mathcal{Q}_{S,t}$: aggregate effective quality

Cross-sector labor wedges, Stone-Geary preferences, Iceberg costs

- Cross-sector labor wedges:

$$w^a_{j,t} = \frac{w^n_{j,t}}{1 + \tau_{w,j}}$$

- Stone-Geary preferences:

$$C_{j,t} = \left(C^a_{j,t} - \bar{a}L_j\right)^{\phi} \left(C^n_{j,t}\right)^{1-\phi}$$

- Iceberg trade costs for agricultural goods:

$$\sum_{k \in \{N,S\}} \tau_{j,k}\, p^a_{k,t}\, c^a_{j,k,t} + C^n_{j,t} = E_{j,t}, \quad \text{where} \quad \tau_{j,k} = \begin{cases} 1 & \text{if } k = j, \\ \tau_g & \text{if } k \neq j. \end{cases}$$

Back

Quantitative model: Agricultural productivity differences in period t

$$\frac{Y^a_{S,t}/L^a_{S,t}}{Y^a_{N,t}/L^a_{N,t}} = \left[\left(\frac{A^a_S}{A^a_N}\right)\left(\frac{Z_S}{Z_N}\right)^{\gamma} \cdot \left(\frac{A^n_S}{A^n_N}\cdot\frac{1+\tau_{w,N}}{1+\tau_{w,S}}\right)^{\alpha+\gamma\varepsilon} \cdot \mathcal{Q}(\lambda_{t-1})^{\alpha} \cdot M_t^{-\gamma}\right]^{\frac{1}{1+\gamma(\varepsilon-1)}}$$

where

$$\mathcal{Q}(\lambda) := \left[\frac{\left(e^{-\delta}q_N\right)^{\theta} + (q_S)^{\theta}}{(q_N)^{\theta} + \left(e^{-\delta}q_S\right)^{\theta}}\right]^{1/\theta} \quad \text{and} \quad M_t := \frac{\tau_g{}^{1-\varepsilon}\, E^a_{N,t}\left(P^a_{N,t}\right)^{\varepsilon-1} + E^a_{S,t}\left(P^a_{S,t}\right)^{\varepsilon-1}}{E^a_{N,t}\left(P^a_{N,t}\right)^{\varepsilon-1} + \tau_g{}^{1-\varepsilon}\, E^a_{S,t}\left(P^a_{S,t}\right)^{\varepsilon-1}}.$$

- Effective quality $\mathcal{Q}(\lambda)$ becomes smooth function, reflecting continuous input type intensity
- M_t: South's market access relative to North's
- Land and Sectoral labor wedge

▸ Back: Baseline ▸ Back: Quantitative

Moscona-Sastry (2025)

- **Measurement:** CABI data on presence of crop pests & pathogens (CPPs) by country
  - For crop k, countries ℓ, ℓ': CPP mismatch$_{k,\ell,\ell'} = 1 - \frac{\#\text{ common CPPs}}{\sqrt{\#\text{ CPPs}_{k,\ell} \cdot \#\text{ CPPs}_{k,\ell'}}} \in [0,1]$

- **Empirics:** country-crop regression with country & crop fixed effects
  - $\log(\text{crop output}_{k,\ell}) = \beta \cdot \text{CPP mismatch with frontier}_{k,\ell} + \chi_\ell + \chi_k + \text{controls}$
  - $\hat{\beta} = -9.71$ under frontier = US for all crops
  - Combined with model, **prod loss of** $9.71/2.46 = 3.95$ **log points per unit of mismatch**

- **Mapping to δ:**
  - Tanzania's area-weighted mean mismatch with US $= 0.367 \Rightarrow \delta = 3.95 \times 0.367 = 1.45$

Back

Campi-Nuvolari (2021) index of IP protection for plant varieties

- Period: 1961–2018.
- Coverage: 104 countries (73 UPOV members in 2018 + 31 with comparable systems)
- Index has five components:
  1. Ratification of UPOV conventions
  2. Farmers' exception (*farmers' right to save seeds*)
  3. Breeders' exception
  4. Protection length
  5. Patent scope
- Index normalized by highest country score

Back

Model fit to untargeted moments

Moment	Model	Data
Agricultural employment share in 2017	0.600	0.691
Share of the pair's agricultural sales revenue in 2017	0.071	0.081
GDP per worker relative to the US in 2017	0.073	0.073
APD_t in 1982	1/120	1/123
APD_t in 2017 relative to 1982	0.75	0.78
the same on the undirected path	1.06	–

Back

Calibration of (a, χ): Basic idea

- Infer quality ratio λ_t from adoption share of South-adapted inputs in model:

$$\omega_t = \frac{(\lambda_t e^{\delta})^{\theta}}{1 + (\lambda_t e^{\delta})^{\theta}} \qquad \Longleftrightarrow \qquad \lambda_t = e^{-\delta} \left[\frac{\omega_t}{1 - \omega_t} \right]^{1/\theta}$$

- **Idea:**

1. **Measure** ω_t at 1970, 1980, 1990, 1998 in six **Green-Revolution** economies (IND, PAK, BGD, IDN, PHL, MEX) from Evenson–Gollin (2003)
2. **Invert** each observed share into an implied state. India: $0.18 \to 0.33 \to 0.48 \to 0.63$
3. **Fit the law**: choose (a, χ) whose path passes closest to implied states of all six economies

$$\lambda_t = \chi - (\chi - \lambda_{-1})(1 - a)^t$$

- Assumption: $m_{S,t} = 1$ during Green-Revolution episode

Back

Inferring λ_{-1} from λ_{2017}

- Initial period $=$ 1982. 2017 is outcome of seven periods of model's own dynamics

- λ_{-1} is inferred from model's law of motion:

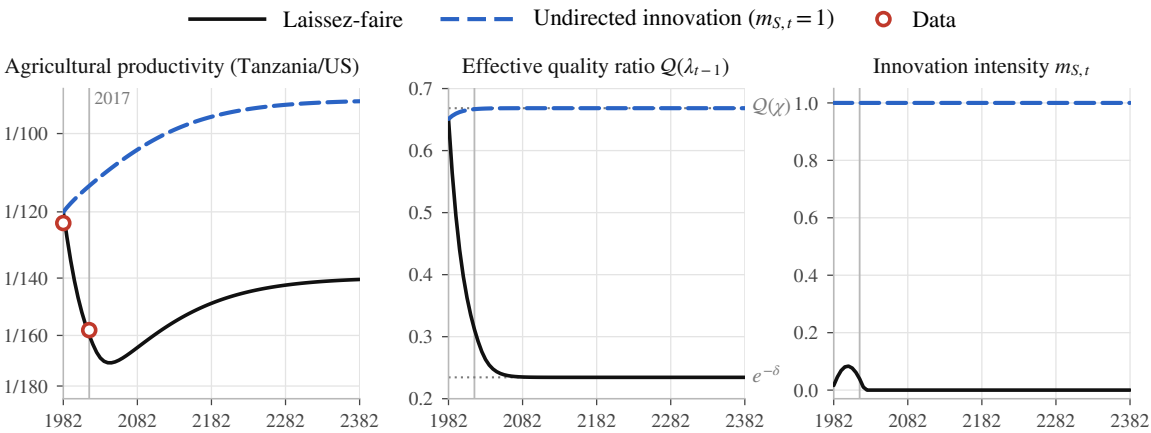
$$F_{2012} \circ \cdots \circ F_{1982}(\lambda_{-1}) = \lambda_{2017}, \qquad F_y(\lambda) = \lambda \exp\left\{m_{S,y}(\lambda)\, g_S(\lambda) - g_N\right\}$$

  - Run law of motion backward from measured 2017 state

  - Innovation intensity is model-implied value: $m_{S,y}$ is what private innovators choose.

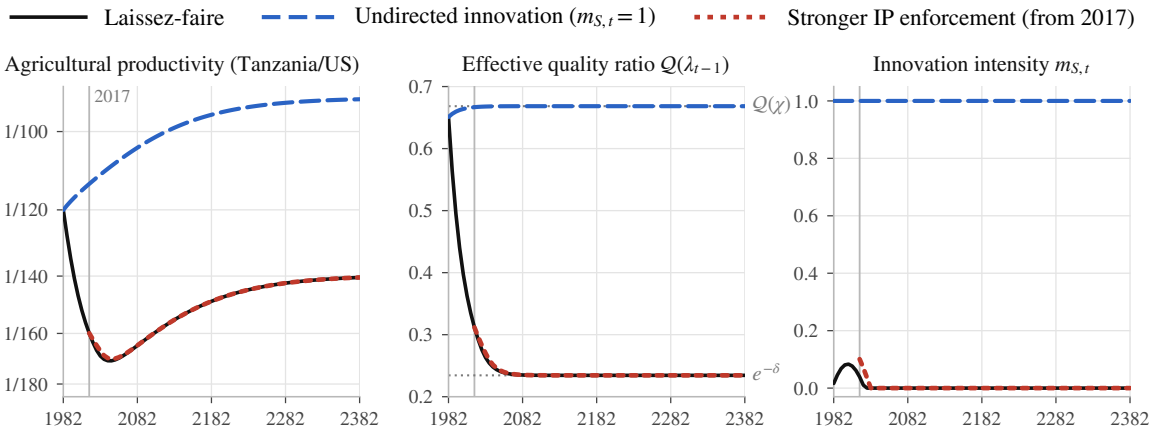
Back

Directed innovation contributes to productivity gaps by 42% in 2017



Back

IP reform does little to narrow agricultural productivity differences



Back

Welfare gains: Equivalent variation in income

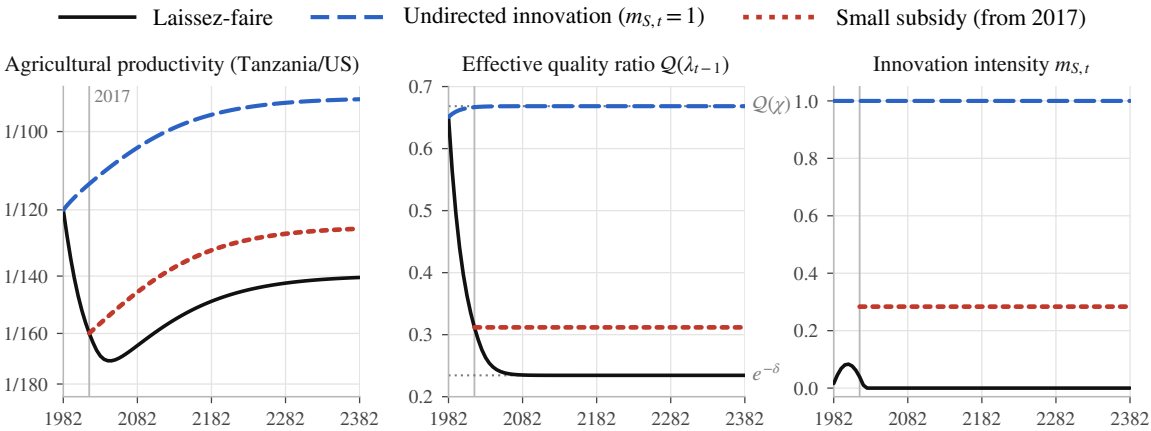
- Welfare gains are measured as equivalent variation in income
- Welfare gains EV_j solve

$$\sum_{t=t_0}^{\infty} \frac{1}{(1+\rho)^{\,t-t_0}}\, u\Big(C_{j,t}^{\mathrm{DE}}\big((1+\mathrm{EV}_j)\,E_{j,t}^{\mathrm{DE}}\big)\Big) = \sum_{t=t_0}^{\infty} \frac{1}{(1+\rho)^{\,t-t_0}}\, u\big(C_{j,t}\big),$$

  - $C_{j,t}^{\mathrm{DE}}(E)$: consumption aggregate that household in country j chooses in period t of laissez-faire path when its income is E, at that path's prices

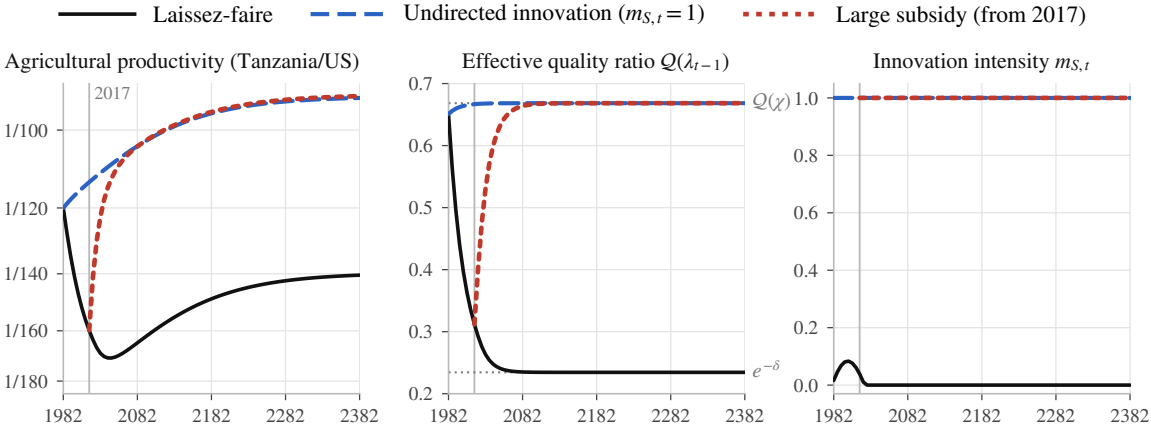
▸ Back: IP ▸ Back: decomposition

Small R&D subsidy narrows productivity differences & raises welfare



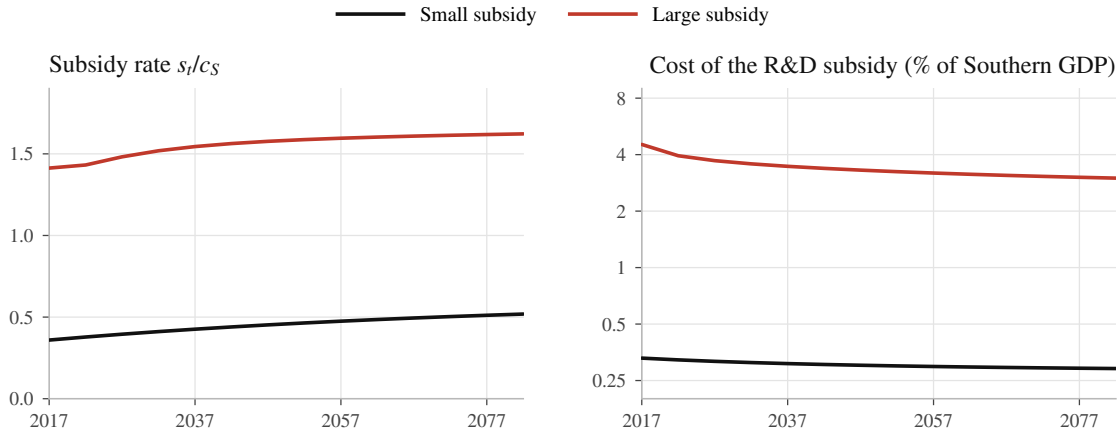
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Large R&D subsidy substantially narrows productivity differences



Back

Cost of R&D subsidy



▸ Back: Small ▸ Back: Large

Environmental heterogeneity & Innovator's tailoring decisions

- Agricultural productivity difference in BGP where only Northern inputs are used:

$$\frac{Y^a_{S,t}/L^a_{S,t}}{Y^a_{N,t}/L^a_{N,t}} = \left[\left(\frac{A^a_S}{A^a_N} \right) \left(\frac{Z_S}{Z_N} \right)^{\gamma} \cdot \left(\frac{A^n_S}{A^n_N} \right)^{\alpha+\gamma\varepsilon} \cdot e^{-\delta\alpha} \cdot \underbrace{\left(\frac{\Psi_S(\theta^*_N)}{\Psi_N(\theta^*_N)} \right)^{\gamma}}_{<1} \right]^{\frac{1}{1+\gamma(\varepsilon-1)}}$$

  - Agricultural productivity differences depend on:

    1. Agricultural fundamentals
    2. Non-agricultural productivity
    3. Macro-environmental mismatch
    4. Micro-environmental mismatch: Tailoring directed more toward North generates larger micro-environmental mismatch in South

▸ Back

Environmental heterogeneity and effectiveness of input in Agriculture

- Each country has land Z_j & land density g_j on environmental chracteristics space $[0, 1]$
  - Total land: $Z_j = \int_0^1 Z_j(i)di$, where $Z_j(i)$: land w/ characteristic i. Land density: $g_j(i) = \frac{Z_j(i)}{Z_j}$
  - *Uniform density:* $g_j(i) = (1/\sigma_j) \cdot \mathbb{1}\{i \in [\nu_j - \sigma_j/2, \nu_j + \sigma_j/2]\}$
  - **South exhibits larger environmental heterogeneity:** $\sigma_S > \sigma_N$ and $\nu_S \neq \nu_N$ (sps. $\nu_N < \nu_S$)

- Effectiveness of intermediate inputs depends on environment to which it is adapted:

$$B_{j,k}(i; \theta_k) = \underbrace{e^{-\delta \mathbb{1}\{k \neq j\}}}_{\text{across-environment}} \cdot \underbrace{B_w(|i - \theta_k|)}_{\text{within-environment}}$$

1. Macro-mismatch $e^{-\delta \mathbb{1}\{k \neq j\}}$: Large-scale differences across environmental types k
2. Micro-mismatch $\underbrace{B_w(|i - \theta_k|)}_{\in [0,1]}$: Distance btw land characteristic i & innovator's target θ_k

   - *Triangular kernel:* $B_w(d) = \max\{0, 1 - d/r\}$; Threshold: $B_w(d) = \mathbb{1}\{d \leq r\}$

Technology in Agriculture with macro- and aggregate micro-mismatch

$$Y_{j,t}^a = A_j^a (L_{j,t}^a)^\beta Z_{j,t}^\gamma \left[\left(e^{-\delta \mathbb{1}\{N \neq j\}} \cdot \Psi_j(\theta_N)^{\frac{\gamma}{\alpha}} \cdot q_{N,t} \right) x_{j,N,t} + \left(e^{-\delta \mathbb{1}\{S \neq j\}} \cdot \Psi_j(\theta_S)^{\frac{\gamma}{\alpha}} \cdot q_{S,t} \right) x_{j,S,t} \right]^\alpha$$

- $\Psi_j(\theta)$: aggregate micro-mismatch from within-country environmental heterogeneity

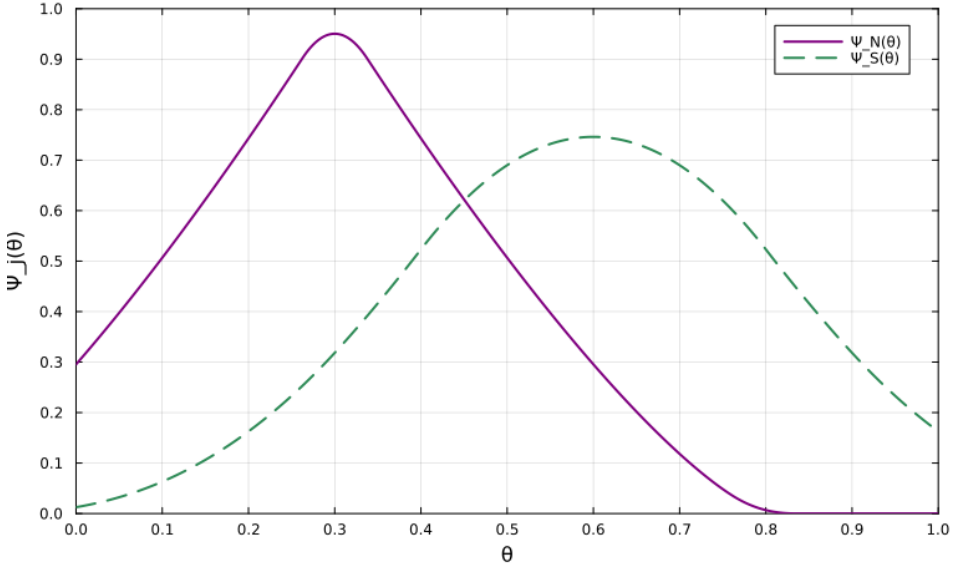
$$\Psi_j(\theta) := \int_0^1 g_j(i) \cdot B_w(|i - \theta|)^{\alpha/\gamma} di$$

Key results:

1. $\Psi_j(\theta) \in [0, 1]$ for all $\theta \in [0, 1]$
2. $\Psi_j(\theta)$ attains unique maximum at $\theta = \nu_j$, denoted Ψ_j^*, and is strictly decreasing in $|\theta - \nu_j|$
3. Maximum value Ψ_j^* is strictly decreasing in σ_j

- $q_{k,t}$: quality of intermediate inputs adapted to environment type k

Graph of aggregate micro-mismatch function Ψ_j



Stage 2: Tailoring decision

- Stage 2: Tailoring decision
  - R&D firm selects θ_k to maximize stage-3 profit $\pi_{k,t}(\theta_k)$, taking aggregate vars as given:

$$\max_{\theta_k \in [0,1]} \quad \alpha(1-\alpha) \sum_{\{j:\tau_t^*(j)=k\}} p_{j,t}^a Y_{j,t}^a(\theta_k)$$

    - R&D firm internalizes effect of θ_k *only* on its own intermediate input demand $x_{j,k,t}(p_{k,t}^x; \theta_k)$

Directed innovation: Optimal tailoring biased toward North under FA

- Stage-2 problem of R&D firm innovating inputs adapted to North under FA regime:

$$\max_{\theta_N \in [0,1]} \; \alpha(1-\alpha)\left[p^a_{N,t} Y^a_{N,t}(\theta_N) + p^a_{S,t} Y^a_{S,t}(\theta_N)\right]$$

- Optimality condition in equilibrium:

$$\underbrace{-\Psi'_N(\theta^*_N)}_{\text{MC for moving right}} = \underbrace{\left[\left(\frac{A^a_S}{A^a_N}\right)\left(\frac{Z_S}{Z_N}\right)^{\gamma} \cdot \left(\frac{A^n_N}{A^n_S}\right)^{\beta} \cdot e^{-\delta\alpha}\right]^{\frac{\varepsilon-1}{1+\gamma(\varepsilon-1)}} \cdot \left(\frac{\Psi_N(\theta^*_N)}{\Psi_S(\theta^*_N)}\right)^{\frac{1}{1+\gamma(\varepsilon-1)}}}_{\text{equilibrium relative weight}} \cdot \underbrace{\Psi'_S(\theta^*_N)}_{\text{MB for moving right}}$$

- **Equilibrium $\theta^*_N \in (\nu_N, \nu_S)$ lies much closer to Northern mode than to Southern mode:**
  1. *Market size:*
  2. *Curvature of Ψ_j:*
  3. *Equilibrium feedback:*

▸ Graph

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